THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A cell that has a nucleus, a cell wall and chloroplasts is likely from: 1

(a) a leaf **(b)** bacterium **(c)** blood **(d)** human cheek

2. In a village, many people fell ill with diarrhoea after a flood. The most probable cause is: 1

(a) mosquito bite **(b)** contaminated drinking water
(c) sunlight **(d)** dust

3. Two wires of equal length and thickness, one nichrome and the other copper, carry the same current. Which gets hotter? 1

(a) nichrome **(b)** copper **(c)** both equally **(d)** neither

4. A cart is pushed with 20 N and friction is 12 N. The net force is: 1

(a) 20 N **(b)** 32 N **(c)** 8 N **(d)** 12 N

5. A brick of weight 20 N has faces of areas 0.1 square metre and 0.04 square metre. The maximum pressure it can exert when resting on a face is: 1

(a) 0.8 Pa **(b)** 200 Pa **(c)** 2 Pa **(d)** 500 Pa

6. When 10 g of salt is dissolved in 100 mL of water, the volume of the solution is hardly more than 100 mL. This is because: 1

(a) particles of salt fit in the gaps between water particles
(b) water evaporates entirely
(c) salt becomes gas
(d) salt disappears

7. Which of the following pairs contains only compounds? 1

(a) O_3 and CO_2 **(b)** N_2 and CH_4
(c) NH_3 and CH_4 **(d)** Fe and CO_2

8. How many grams of sugar are in 250 g of a 12 per cent sugar solution? 1

(a) 12 g **(b)** 25 g **(c)** 30 g **(d)** 3 g

9. The angle between the incident and the reflected ray is 80 degrees. The angle of incidence is: 1

(a) 40 degrees **(b)** 10 degrees **(c)** 160 degrees **(d)** 80 degrees

10. At noon the shadow of a stick is directly towards the north (in India). The Sun then is: 1

(a) below the horizon **(b)** to the north
(c) to the south of the observer **(d)** overhead

11. A sudden increase in the population of rabbits in a grassland may occur if: 1

(a) foxes are removed **(b)** owls are added
(c) rains are heavy **(d)** grass is removed

12. Venus has a surface temperature of about 460 degrees Celsius mainly due to: 1

(a) closeness to the Moon **(b)** its magnetic field
(c) a thick CO2 atmosphere **(d)** its oceans

13. For a study with the dependent variable 'height of a plant', the most suitable unit is: 1

(a) degree C **(b)** cm **(c)** s **(d)** kg

14. A student gets a 400x total magnification using a 40x objective lens. The eyepiece lens is: 1

(a) 10x **(b)** 20x **(c)** 5x **(d)** 100x

15. Vaccines are more effective when given: 1

(a) after the symptoms appear **(b)** before exposure to disease
(c) only to adults **(d)** only after recovery

16. A scrap yard crane drops the iron when: 1

(a) the current is on **(b)** the current is switched off
(c) the iron is heavy **(d)** the wire is long

17. A charged balloon attracts a neutral wall because: 1

(a) of gravity only **(b)** it induces opposite charge near it
(c) the balloon is heavy **(d)** the wall is a magnet

18. A sealed bag of chips puffs up on a hilltop because: 1

(a) chips expand **(b)** outside pressure is higher
(c) outside pressure is lower than inside **(d)** air inside cools

19. A gas can be turned into a liquid if it is cooled below its: 1

(a) boiling point (condensation point) **(b)** melting point only
(c) room temperature **(d)** freezing point only

20. Which of the following is NOT an alloy? 1

(a) copper **(b)** steel **(c)** brass **(d)** bronze

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Why is the inner core solid though the temperature is very high? 2

22. State two limitations of using only the senses for observation. 2

23. Differentiate between prokaryotic and eukaryotic cells (one point each). 2

24. How are vaccines different from antibiotics? 2

25. Why does an electromagnet pick up more pins when the number of turns is increased? 2

26. Why do athletes use spikes on their shoes? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Explain how an MCB works. What is the advantage over a fuse? 3

28. Explain why the weight of an astronaut changes on the Moon but not the mass. 3

29. Differentiate between a cyclone, a tornado and a thunderstorm (one point each). 3

30. Explain the change in arrangement of particles when water vapour changes to ice. 3

31. Give three reasons why gold and silver are used in jewellery. 3

OR

Describe the steps for obtaining crystals of copper sulphate from its saturated solution. 3

32. Why does the driver of a vehicle prefer convex rear-view mirrors? Mention any drawback. 3

OR

Compare the Gregorian, Islamic and Hindu calendars on basis of the Sun, Moon and year length. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Describe the safety measures a community should take before, during and after a cyclone. 5

34. Describe how pressure and temperature can be used to liquefy gases and give three uses. 5

35. Discuss the use of metals and non-metals in the human body, daily life and industry with five examples. 5

36. Explain the changes that occur when a saturated solution of KNO3 at 60 degrees Celsius is cooled to 20 degrees Celsius. 5

OR

A man 1.8 m tall stands in front of a plane mirror. Explain the minimum size of mirror and where to place it to see his full image. Show the geometry in words. 5

37. A traveller finds the Pole Star 25 degrees above the northern horizon. Find his latitude and describe how one can locate the Pole Star with the Big Dipper. 5

OR

Compare a pond and a grassland ecosystem on the basis of producers, consumers, decomposers, abiotic factors, and give one food chain for each. 5

38. Explain the causes and effects of ozone depletion, mention the gases responsible, and the steps taken by the world to repair the layer. 5

OR

Marks of a pendulum experiment: length (cm) 20, 40, 60 and time period (s) 0.9, 1.3, 1.6. Describe the relationship and a conclusion. State two variables to keep constant. 5

END OF QUESTION PAPER — Total: 80 Marks

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